Forest Ecosystems 11 (2024) 100233

Contents lists available at ScienceDirect

Forest Ecosystems

journal homepage: www.keaipublishing.com/cn/journals/forest-ecosystems

A bibliometric analysis using machine learning to track paradigm shifts and analytical advances in forest ecology and forestry journal publications from 2010 to 2022

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ARTICLE INFO

Keywords:
Forest ecology
Forestry
R software
Structural topic modelling
Machine learning
Publication

ABSTRACT

Forest habitats are critical for biodiversity, ecosystem services, human livelihoods, and well-being. Capacity to conduct theoretical and applied forest ecology research addressing direct (e.g., deforestation) and indirect (e.g., climate change) anthropogenic pressures has benefited considerably from new field- and statistical-techniques. We used machine learning and bibliometric structural topic modelling to identify 20 latent topics comprising four principal fields from a corpus of 16,952 forest ecology/forestry articles published in eight ecology and five forestry journals between 2010 and 2022. Articles published per year increased from 820 in 2010 to 2,354 in 2021, shifting toward more applied topics. Publications from China and some countries in North America and Europe dominated, with relatively fewer articles from some countries in West and Central Africa and West Asia, despite globally important forest resources. Most study sites were in some countries in North America, Central Asia, and South America, and Australia. Articles utilizing R statistical software predominated, increasing from 29.5% in 2010 to 71.4% in 2022. The most frequently used packages included lme4, vegan, nlme, MuMIn, ggplot2, car, MASS, mgcv, multcomp and raster. R was more often used in forest ecology than applied forestry articles. R software offers advantages in script and workflow-sharing compared to other statistical packages. Our findings demonstrate that the disciplines of forest ecology/forestry are expanding both in number and scope, aided by more sophisticated statistical tools, to tackle the challenges of redressing forest habitat loss and the socio-economic impacts of deforestation.

1. Introduction

According to a recent report by the Food and Agriculture Organization of the United Nations and the United Nations Environment Program (FAO and UNEP, 2020), forests currently cover c. 31% of the earth's land surface, amounting to c. 4.06 billion hectares, of which around half remains relatively intact, and one third as primary forest where there are no visible indications of human activities impacting ecological processes. These forests are, however, not distributed equally, with more than half existing in only five countries (the Russian Federation, Brazil, Canada, the United States of America, and China), with two-thirds occurring over just ten countries.

Globally, forests absorb around 16 billion metric tonnes of CO_2 per year while emitting an average of 8.1 billion metric tonnes, for a net annual sequestration of 7.6 billion metric tonnes of CO_2 (Harris et al., 2021). Humans have removed almost half of Earth's natural forests (Crowther et al., 2015; Olagunju, 2015), leading to an annual loss of carbon absorption capacity ranging from 0.9 to 2.3 billion tonnes of CO_2 (about 15% of annual human carbon emissions) due to deforestation (Friedlingstein et al., 2020). Forests also play a major global role in numerous ecosystem functions and services, including nutrient cycling, nitrogen fixation, water purification, biomass production, habitat provisioning services, pollination, seed dispersal, resistance to windstorms, fire regulation and mitigation, regulation of native pests and invasive insects, and provide socio-economic ecosystem services in relation to forest type, structure, and diversity (Brockerhoff et al., 2017).

As habitats, forested regions support around 80% of terrestrial plant and animal biodiversity (FAO and UNEP, 2020), including an estimated

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https://doi.org/10.1016/j.fecs.2024.100233
Received 9 February 2024; Received in revised form 28 July 2024; Accepted 28 July 2024
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60,082 tree species (BGCI, 2023). Of these, 20,334 tree species are included in the IUCN Red List of Threatened Species (IUCN, 2019a), 8,056 of which are assessed as globally threatened, and over 1,400 are assessed as critically endangered and in urgent need of conservation action. In addition, around 60% of the world's known 391,000 species of vascular plants (of which about 94% are flowering plants) are found in tropical forests (Burley, 2002), along with the substantial majority of fungi species (Willis, 2018). Of 70,000 known and described vertebrate species (IUCN, 2019a), forests provide habitats for c. 5,000 amphibian species (80% of all known species), c. 7,500 bird species (75% of all species) and more than 3,700 mammals (68% of all species) (Vié et al., 2009). Furthermore, the vast majority of those 1.3 million invertebrate species have been described live in forests (Ødegaard, 2000). Within forest communities, species assemblages have various inter-dependencies, supporting functions such as pollination, seed dispersal, and decomposition by saproxylic beetles, detritivores, and fungi. In turn, these processes feedback to determine the distribution patterns of trees and forest structural complexity (Audino et al., 2017).

Globally, forests are also important for human livelihoods, with the FAO and UNEP (FAO and UNEP, 2020) report estimating that forests provide more than 86 million direct green jobs, with another 880 million people using forests for the collection of fuel wood or to produce charcoal. Other non-fuel wood products (NFWPs) and non-timber forest products (NTFPs) are collected for personal use or sale, with varying degrees of sustainability (Pandey et al., 2016; Derebe et al., 2023). Over 90% of people living in extreme poverty are dependent on forests for at least part of their livelihoods. These are often indigenous people, who manage c. 28% of the world's land surface, including many ecologically important forest regions (FAO and UNEP, 2020).

Beyond subsistence, non-consumptive uses of forest biodiversity include recreation and tourism, with an estimated 8 billion visits to protected forest areas each year (Balmford et al., 2009; FAO and UNEP, 2020). The attraction of forest habitat is so imprinted on the human psyche that time spent in forests and wooded urban green spaces even brings recognized health benefits (Oh et al., 2017; Grilli and Sacchelli, 2020). To date, however, only 18% of the world's forest area, around 700 million hectares, fall within legally established protected areas such as national parks, conservation areas and game reserves (IUCN categories I–IV). This marginally exceeds Aichi Biodiversity Target 11 (to protect at least 17% of terrestrial area by 2020 (IUCN, 2019b). South America protects the highest proportion of its forests (31%), with just 5% of European forests in protected areas. Importantly, these areas are not yet fully representative of the global diversity of forest ecosystems.

Despite this importance, forests are being lost at an alarming rate, with c. 420 million hectares converted to other land uses since 1990, although this rate of reforestation decreased to c. 10 million hectares per year between 2015 and 2020, down from c. 16 million hectares per year during the 1990s (FAO and UNEP, 2020). Losses are primarily due to agricultural expansion, with cattle ranching and the cultivation of soya beans and palm oil contributing 40% of tropical deforestation between 2000 and 2010, with a further 33% due to local subsistence agriculture (FAO and UNEP, 2020).

It is therefore imperative to better understand and conserve forest ecosystems, and to develop sustainable forestry and reafforestation practices (Xie et al., 2023), and, to these ends, huge global effort is invested into research that falls into two inter-related fundamental domains, forest ecology and forestry practice. Forest ecology involves theoretical and conceptual studies aimed at elucidating the mechanisms underlying forest ecosystems, such as studies of diversity and complexity (Mori et al., 2017), community interactions (Kunstler et al., 2012), energy flux (Harris et al., 2021), hydrology (Zhang et al., 2017), pollination (Hanula et al., 2016), seed dispersal (Lei et al., 2021), etc. In comparison, practical forestry concentrates on the delivery of ecosystem services from forest habitats, such as timber production/harvesting (Bont et al., 2022), tree planting (Brancalion and Holl, 2020), NTFPs, wood fuel and NWFPs (Sheppard et al., 2020), and silviculture (Ashton and Kelty, 2018), etc.

Within both of these sub-disciplines, advancing technology and research tools over recent decades have revolutionized the types of study that can be conducted; for instance, the use of ariel drones (Zhang et al., 2016), remote sensing (Lechner et al., 2020), and LiDAR (Beland et al., 2019), as well as genomics (Grattapaglia et al., 2018) and epigenetics (Sow et al., 2018; Amaral et al., 2020) were used to accelerate the adaptive breeding of trees. New statistical techniques have also further transformed the sophistication of forest ecology studies, vastly expanding the understanding of key topics. Noteworthy in this regard is the way in which the R programming language has supplanted former statistical software (Lai et al., 2019), aided by the wealth of opensource packages developed for this application (Crawley, 2012; Oyana, 2020). However, the full potential of R for integrating comprehensive statistical analyses in forest ecology and especially applied practical forestry is yet to be fully realized (Hackenberger, 2020; Mehtatalo and Lappi, 2020; Atkins et al., 2022).

The ability to characterize and analyze bibliometric trends in forest ecology and forestry, to support research synergies and strategic planning, has advanced considerably in recent years due to greater computer processing power, advances in machine learning, and the availability of large, digitised text corpora online, which has enabled a rapid increase in the use of automated content analysis computational methods (Donthu et al., 2021). For example, a recent forestry research study (Nummelin et al., 2021) used topic modelling to identify 160 commercial forestry topics. Similarly, bibliometric methods have been used to quantify trends in forest ecosystem services research (Aznar-Sánchez et al., 2018). Nevertheless, a substantial knowledge deficit remains concerning general trends in forest ecology/forestry research.

To address this deficit, here we used structural topic modelling (STM) to examine a total of 16,952 articles published in 13 journals between January 2010 and March 2022, aiming to detect trends among topics dominating forest ecology and forestry publications. Therein, we gave particular attention to the increasing use of statistical procedures, particularly those using the R domain. We specifically addressed five questions: (1) What were the key themes investigated by studies over the entire corpus? (2) How did the representation of each theme change over the past 12 years? (3) Which countries were spearheading key research topics? (4) What was the trend in the use of R in analyses? And (5) what R packages were used most frequently for forest ecology? We then use our results to examine how changing trends in forest ecology and forestry align with globally important demands on forest ecosystems.

2. Materials and methods

2.1. Data collection

We selected the top five journals covering forest ecology and forestry based on their 5-year impact factor and downloaded the full text of each paper published between 1st January, 2010 and 31st March, 2022. We also selected the top eight multidisciplinary biology and ecology journals featuring forest ecology/forestry articles published between 1st January, 2010 and 31st March, 2022 (Table S1), and conducted a literature search using the ISI Web of Science, applying the keywords [forest* AND eco*] OR [forest* AND Manag*] OR [forest*] and the Publication/Source Titles [Nature] OR [Science] OR [Nature Climate Change] OR [Nature Ecology and Evolution] OR journals listed in Table S1. This search yielded 17,389 publications, of which 16,952 articles were downloaded after excluding reviews and comments and the single main topic covered by each was established by analysing their abstracts using structural topic models (STM; see below). Full article texts were examined manually to ascertain the statistical software used in each study (Fig. S1). Our corpus omitted articles published in local journals because of the language barriers, whereas the substantial majority of research is published in English (Amano et al., 2016). Furthermore, we limited our research to the top journals for forest ecology/forestry to focus on the most impactful studies published. The purpose of this form of text analysis was to identify major

themes in the literature, therefore adding more journal sources would be unlikely to change these broad patterns.

2.2. Topic models

To prepare this literature corpus for analysis, we used the 'textProcessor' function in the R package 'stm' (Roberts et al., 2019) to remove any numbers, articles (e.g., a, the, these), conjunctives (e.g., and, the, or), prepositions (e.g., in, under, on, through) personal pronouns (e.g., we, our, they, I, you) and punctuation from text extracted from all article abstracts. We also removed adjectives, adverbs, verbs, and neutral terms (e.g., year, day, study, research) and restricted minimum word length to at least 3 letters. We further restricted the keyword list by removing uninformative words (e.g., results, suggested, different, similar). Finally, we removed 'rare' words that appeared in fewer than 1% of articles and 'common' words appearing in more than 85% of articles, using the 'prepDocuments' function, as these also provide very little informative content (Westgate et al., 2015).

To determine the main topics in our corpus of article abstracts, we conducted structural topic modelling (STM) using the 'stm' package in R (Roberts et al., 2019). When fitting an STM to a corpus a prerequisite number of topics must be specified. To determine the number of latent topics represented in our corpus, we applied iterative topic model solutions of between 5 and 100 topics at increments of 5 by using the 'searchK()' function in the 'stm' package. Both cross-validation likelihood and semantic coherence were inspected to ascertain how many latent topics occurred across our dataset. Cross-validation likelihood was used to provide the fit of each solution in a hold-out sample (i.e., a random sample from a dataset) withheld from the model training data, which was then applied to give an unbiased assessment of model performance on new data. Semantic coherence was used to provide a metric across various solutions (Aronoff and Anshen, 2017). This process is akin to visual inspection of a scree plot. When considering both of these criteria, the optimal solution for our dataset was 20 topics; a number that was sufficient to provide a good overview of the corpus while not being too large to interpret (Andrew et al., 2022) (Fig. S2). We used the 'labeltopics()' function in the 'stm' package to generate a set of keywords to provide us with a basis to label each topic. These topic definitions were verified using the findThoughts() function in R 'stm' package.

To understand how these topics co-occurred, we first calculated the topic probability distribution of each reference paper using the 'make.dt()' function in the 'stm' package. The correlation coefficient among topics was also calculated based on the probability distribution of each topic. To reduce the influence of uneven sampling (e.g., differences in abstract length, publication year, and publication rate) among journals, we employed a normalised journal by-topic matrix and used Bray-

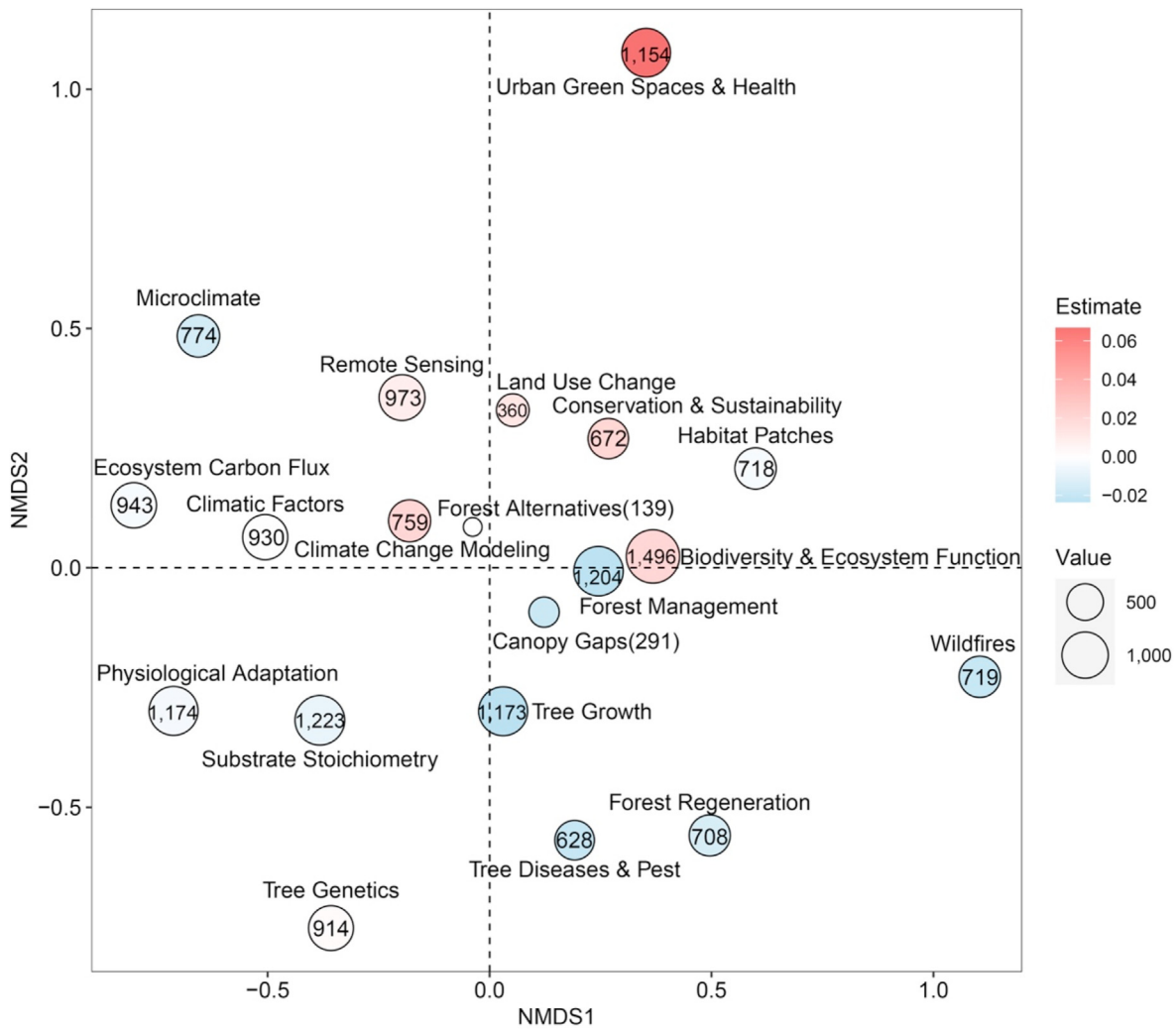

Fig. 1. The position of 20 latent forest ecology/forestry topics within overall topic space, based on 16,952 article abstracts published between 2010 and 2022. Circle size is proportional to topic occurrence frequency in the corpus; colour corresponds to topic temporal trends. The total "topic space" was occupied by different topics, according to their NMDS scores. Topics near the center represent those that lie at the core of forest ecology/forestry, while those on the periphery represent frontier topics.

Curtis dissimilarity to estimate the distance between journals. We performed non-metric multidimensional scaling (NMDS; which finds both a non-parametric monotopic relationship between the dissimilarities in the item-item matrix and the Euclidean distances between items, and the location of each item in the low-dimensional space) on this distance matrix to compare similarity between these topics among journals (McCallen et al., 2019). The total 'topic space' was occupied by different topics, according to their NMDS scores. Topics near the centre represented the core of forest ecology/forestry, while those on the periphery represented frontier topics.

To examine how similar topics were to each other, we used Ward's minimum variance method on the squared value of the dissimilarities to build a matrix of co-occurrence for each topic, starting with each of the topics in its own cluster. The cluster number was reduced gradually until every topic was unified in a single cluster, creating a hierarchical classification tree (Ward, 1963). Topic correlation was also analysed using the 'topicCorr' function to estimate relative topic proportions, along with applying the 'pheatmap' function in the 'pheatmap' package to draw these topics as heatmaps.

2.3. Temporal trends among topics

We used the 'estimateEffect' function in the 'stm' package (Roberts et al., 2019) in R to reveal how the prevalence of topics in the corpus changed through time. We treated year as a continuous predictor variable (Roberts et al., 2019) plotted against each topic weight response variable. We divided our 12-year period into 4 intervals, each of 3 years, and calculated the number of articles relating to different topics in each. We used a Sankey plot to display temporal trends among topics using the 'ggplot2' and 'ggalluvial' packages in R.

2.4. Geographical distribution

The country to which each first author was affiliated (i.e., their institutional address) and the frequency distribution of countries represented was deduced using the 'biblioAnalysis' function in the 'bibliometrix' package (Aria and Cuccurullo, 2017) in R. The study site referred to in each article was collected manually, in order to understand where 'hotspot' areas for forest ecology research were located, and to observe trends over time. We plotted maps using the 'maps' and 'ggplot2' packages. We also used linear regression models to investigate if there was any relationship between the area of forest coverage (independent variable; data were obtained from the Food and Agriculture Organization of the United Nations database, https://www.fao.org/faostat/en/#data/RL) in the country where the publication (response variable) was produced and the number of publications originating from that country.

2.5. Trends in statistical methodologies

To detect trends in the statistical analysis methodologies used in these forestry ecology/forestry publications, we manually identified the statistical software used in each article (because the multiple purposes for which statistical software was employed could not be detected accurately by automatic analysis software). We also noted which R packages were cited most frequently in the corpus (above a threshold of being used in $\geq$80 articles). We then analysed how the use of statistical software and R packages changed from 2012 to 2022.

3. Results

3.1. Major topics in forest ecology/forestry

We defined 20 predominant latent topics among our total corpus of 16,952 research articles. These ranged from traditional (e.g., Tree growth, Tree diseases & pests, Forest alternatives) to contemporary (e.g., Conservation and sustainability, Land use change), from microscale (Microclimate, Tree genetics) to macroscale (Biodiversity & ecosystem function, Climate change modeling), and from the fundamental (e.g., Forest regeneration, Canopy gaps) to applied (e.g., Climatic factors, Urban green spaces & health) (Fig. 1, Table S2). Across the study period, the topic that occurred most frequently was Biodiversity & ecosystem function (8.82%), followed by Substrate stoichiometry (7.21%, e.g., nutrient cycles), Forest management (7.10%), Physiological adaptation (6.93%), and Tree growth (6.92%).

From our mapping of forest ecology/forestry topic space, as determined from the topic co-occurrence matrix, we found that traditional topics, such as Forest alternatives, Forest management, and Canopy gaps, were positioned toward the centre of the topic space (Fig. 1), whereas more contemporary topics, such as Urban green spaces & health, Ecosystem carbon flux, and Tree genetics, were positioned toward the periphery.

The 20 topic themes grouped into four clear subject field subdivisions: (i) Forest growth, (ii) Forest management, (iii) Forest use, and (iv) Forest monitoring (Fig. 2, Tables S2, S3). Some of these 20 topic themes correlated positively with each other, such as Conservation & sustainability with Forest alternatives; Climatic factors with Climate change modelling, and Conservation & sustainability with Land use change (Fig. 2, S3, Tables S2, S3).

3.2. Temporal trends in themes of forest ecology

Overall, applied (e.g., Urban green spaces & health), macroscale (e.g., Biodiversity & ecosystem function), and contemporary (e.g., Climate change modelling) topics became more prevalent over the 12-year study period, while traditional (e.g., Forest regeneration), microscale (e.g., Tree growth), and classical (e.g., Tree diseases & pest) topics exhibited a negative trend (Fig. 2). Over the study period, the major focus of research shifted from more applied topics, such as Forest regeneration and maintenance mechanisms, toward more conceptual studies on broader topics, such as Forest ecosystem function and health benefits.

The number of publications produced each year increased substantially over these 12 years, from 820 articles published in 2010 to 2,354 articles published in 2021 (the last complete year in the corpus). However, this increase did not apply across all topics (Fig. 3). The rank of macroscale (e.g., Biodiversity & ecosystem function) and applied topics (e.g., Conservation & sustainability, Urban green spaces & health, Climate change modelling) exhibited a sustained increase, while classical topics (e.g., Forest management, Tree growth, Canopy gaps) underwent a decrease in representation, whereas microscale topics, such as Tree genetics, exhibited significant inter-annual variation.

3.3. Geographical distribution of studies

The greatest proportion of the literature we examined was published by first authors based in the USA (25.3%), followed by China (15.6%), Canada (6.4%), Germany (4.8%), and Australia (4.7%) (Fig. 4). In contrast, some developing countries (e.g., Laos, Angola, and the Central African Republic) had comparatively little footprint, while other countries were entirely absent from the corpus (e.g., Western and Central African nations, Central American nations, and island countries, such as Niue and Samoa). An increased number of articles were published by authors based in countries with greater forest coverage, such as the USA, China, Canada, Australia, and Brazil (Fig. S4). In contrast, the Russian Federation, which has the greatest forest coverage, contributed disproportionately fewer publications on forest ecology/forestry. Overall, the field study sites represented in our corpus were, however, distributed globally, although there were 'hotspot' areas located in some countries in North and South America, Asia, and Australia (Fig. S5).

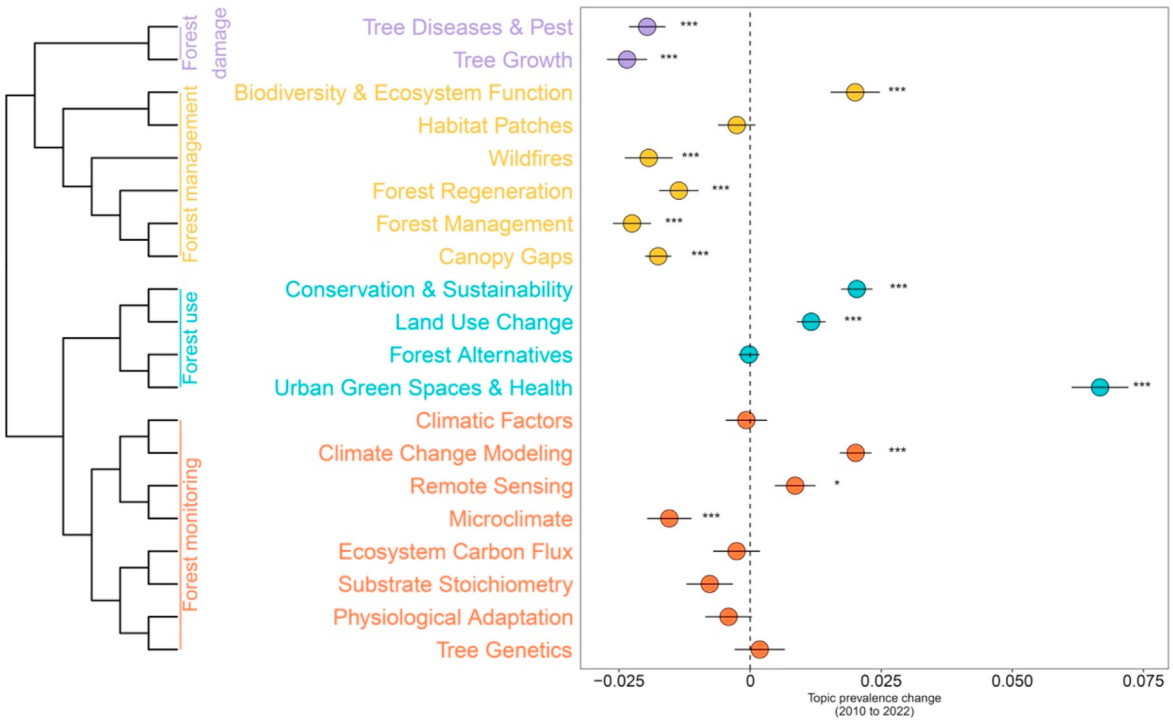

Fig. 2. Change in topic prevalence within the corpus between 2010 and 2022. The position of points on the *x*-axis represents the overall change in the relative representation of each topic over time relative to zero change. Error bars give 95% confidence intervals. The dendrogram to the left of the graph shows the way in which topics cluster to define the four major sub-discipline fields (*** indicates $P < 0.001$, ** indicates $P < 0.05$, * indicates $P < 0.01$).

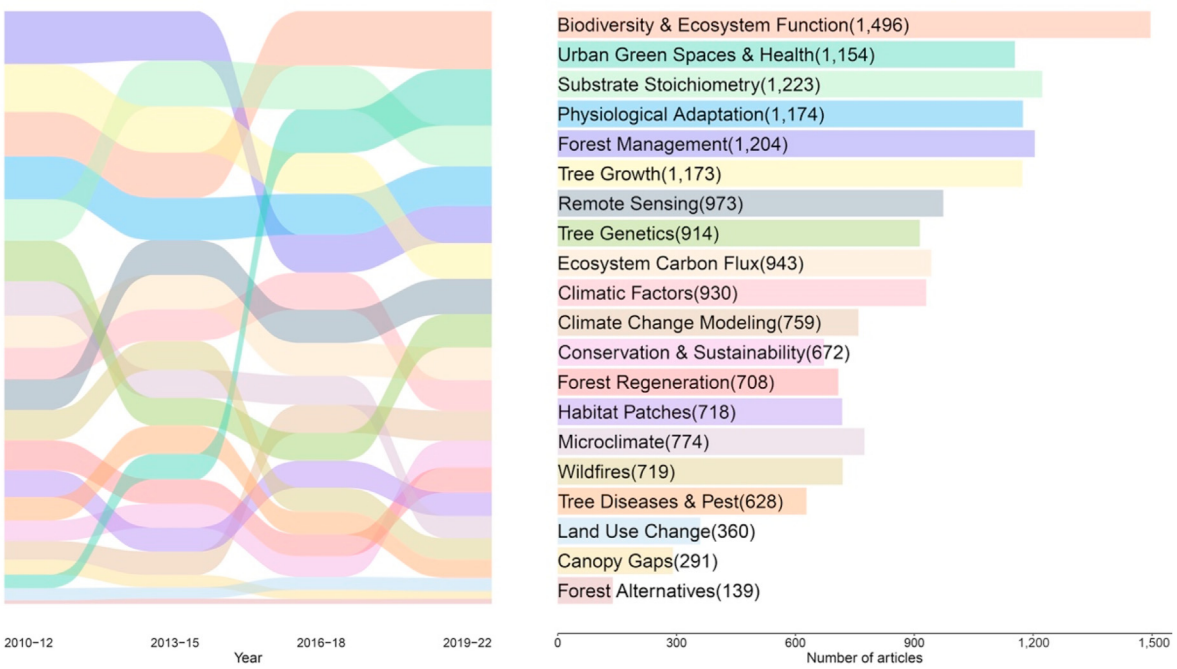

Fig. 3. Sankey plot showing changes in the ranked relative prevalence of the 20 topics in the corpus from 2010 to 2022.

3.4. Changes in statistical methodologies employed

Among our total corpus of 16,952 research articles, 7,059 articles (41.6%) explicitly listed R as the statistical software used for analyses, followed by SPSS (1,669 articles, 9.8%), and SAS (1,284 articles, 7.6%). All journals exhibited a consistent increase in the percentage of articles using R during the study period (Figs. S6, S7), which increased linearly across the time series from 29.5% in 2010 to 71.4% in 2022 (Figs. 5, S7). From articles collated from the eight top multidisciplinary/ecology journals, those that had a particularly high percentage of R use included *Nature Ecology and Evolution* (1st position with 78.4%) (Fig. S8), *Nature Communications* (2nd with 66.0%), *Ecology Letters* (3rd with 64.1%), *Science* (4th with 57.2%), and *Science Advances* (5th with 55.2%). In contrast, articles in the five dedicated forest ecology/forestry journals

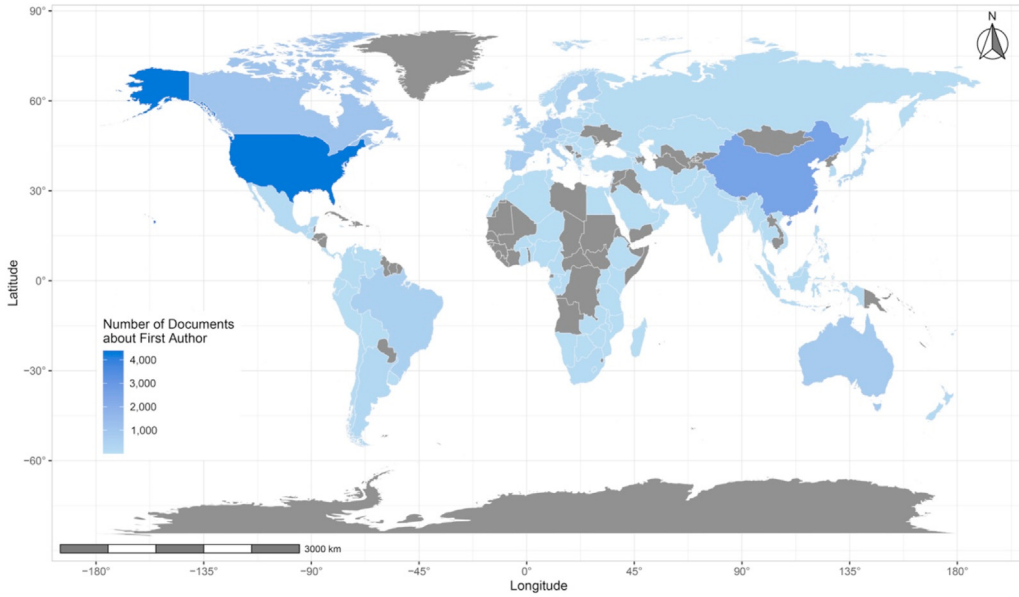

Fig. 4. Global map illustrating the geographic location where the authors of the 16,952 articles in the corpus were located.

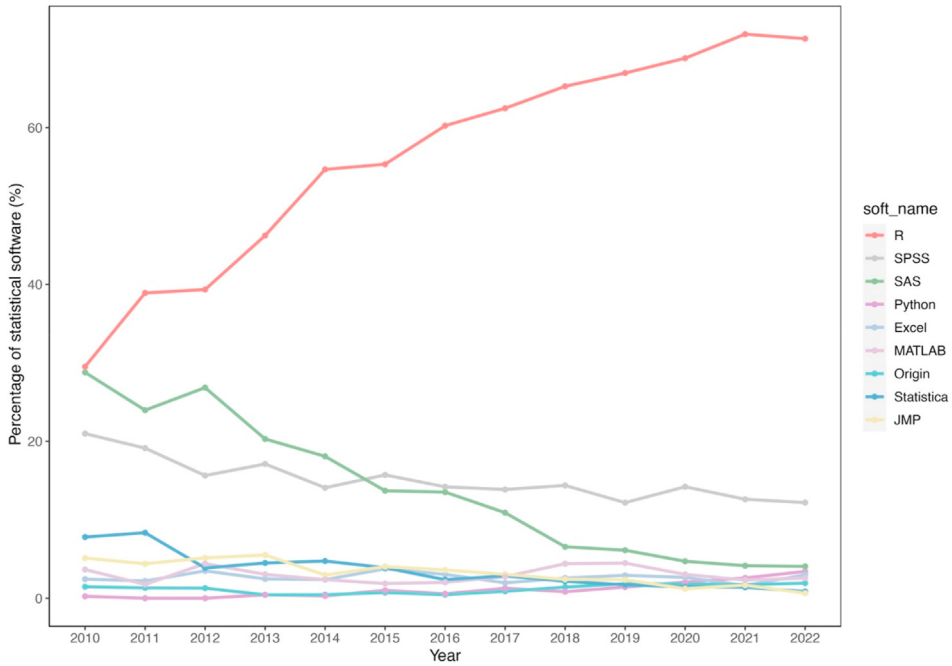

Fig. 5. Graph plotting changes in the relative percentage use of the statistical software cited in the 16,952 articles from the 13 journals between January 2010 and March 2022, noting the ascendence of the R programming language.

had a relative lower percentage of reported R use, led by *Forest Ecosystems* in overall 7th position (52.4%), with *Urban Forestry and Urban Greening* in 13th and last position with just 20.7%. Nevertheless, even within these more focused journals, R was still more popular than other statistical software. Furthermore, R use was dominant across all 20 topic themes, except for Urban green spaces & health (Fig. S9). SPSS and SAS ranked 2nd (14.4%) and 3rd (11.0%), respectively, among other statistical software; only 13.9% of studies did not use one of these top three statistical packages.

3.5. The use of specific R packages

A total of 1,387 R packages were used across these articles, and 17 major R packages featured in ≥80 articles (Fig. 6). These packages ranked (1) 'lme4' (19.8%; used to fit and analyze linear mixed models), (2) 'vegan' (17.8%; used for multivariate analysis in community ecology), (3) 'nlme' (14.5%; used to fit linear and nonlinear mixed models), and (4) 'MuMIn' (8.3%; used to perform model selection and model averaging). The top two packages were used far more often than any other packages. The 10 most frequently used packages all applied to model simulation, except for the package 'ggplot2', which provides a graphics function (Table S5).

Either 'lme4' or 'vegan', were the primary R packages used by articles in 11 of the 13 journals, with *Nature Climate Change* and *Agricultural and Forest Meteorology* as the only exceptions. Articles in six of the eight top multidisciplinary/ecology journals (i.e., *Nature*, *Science*, *Nature Climate*

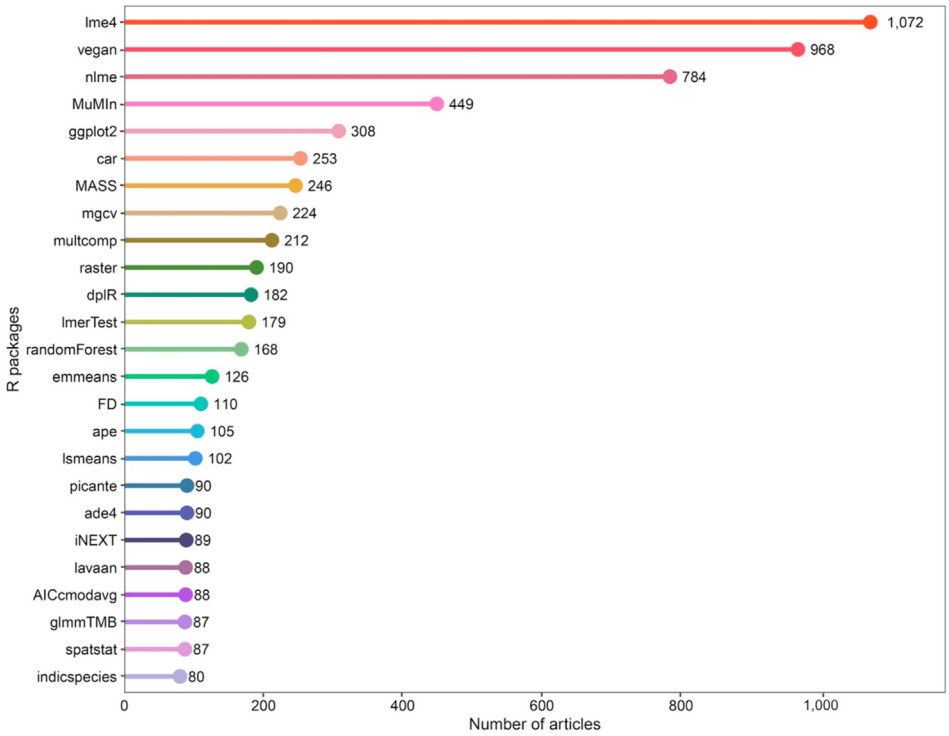

Fig. 6. The R packages (above a threshold of ≥80 articles) cited most frequently in the 16,952 articles from the 13 journals between January 2010 and March 2022.

Change, *Nature Communications*, *Science Advances*, and *Proceedings of the National Academy of Sciences*), also used 'raster', which performs geographic data analysis and modelling. For the five forestry journals, 'lme3', 'vegan', or 'nlme' were used most commonly for modelling and community analysis. The 'nlme' package was used frequently among all journals, except for studies in *Nature Ecology and Evolution* (Table S6).

At the topic level, the 'nlme', 'lme4', and 'vegan' packages were represented among studies covering all 20 latent topics, with 'MuMIn' used in 19/20 topics, with the exception of articles on the topic Forest alternatives (Table 1, S7). The 'dplR' package was used among 15/20 topics. The topic of Physiological adaptation also specifically used the 'fitplc' package (2.9%), which fits Weibull or sigmoidal models to the percentage loss of conductivity (plc) curves as a function of plant water potential, computes confidence intervals of parameter estimates and predictions with bootstrap or parametric methods, as well as and providing convenient plotting functions. The topic of Biodiversity and ecosystem function specifically used the 'FD' package (2.8%), which measures functional diversity (FD) from multiple traits and provides other tools for functional ecology. The 'ggplot2' package, used to display data visually, featured among the top five packages used by five topics (Climatic factors, Tree genetics, Forest alternatives, Tree growth and Microclimate). Table S7 presents the use of statistical software and all R packages by latent topics, ranked from greatest to least use.

4. Discussion

4.1. Temporal trends in forest ecology/forestry topics

Given the vital importance of forests for global ecosystem function and biodiversity, as well as for ecosystem services supporting livelihoods

Table 1
The top 5 R packages used per topic across the corpus (2010 to March 2022).

No.	Title	Package 1	Package 2	Package 3	Package 4	Package 5
1	Biodiversity & ecosystem function	vegan	lme4	nlme	MuMIn	FD
2	Canopy gaps	lme4	nlme	vegan	car	MuMIn
3	Climate change modelling	dplR	nlme	lme4	raster	treeclim
4	Climatic factors	lme4	nlme	mgcv	ggplot2	MuMIn
5	Conservation & sustainability	vegan	lme4	raster	MuMIn	nlme
6	Ecosystem carbon flux	lme4	REddyProc	MuMIn	nlme	car
7	Forest alternatives	lme4	nlme	ggplot2	quantreg	vegan
8	Forest management	vegan	lme4	nlme	MuMIn	MASS
9	Forest regeneration	lme4	nlme	vegan	MuMIn	car
10	Habitat patches	lme4	vegan	MuMIn	nlme	mgcv
11	Land use change	vegan	raster	MuMIn	nlme	lme4
12	Microclimate	lme4	nlme	ggplot2	REddyProc	mgcv
13	Physiological adaptation	nlme	lme4	dplR	fitplc	vegan
14	Remote sensing	randomForest	raster	caret	nlme	vegan
15	Substrate stoichiometry	vegan	lme4	nlme	MuMIn	multcomp
16	Tree diseases & pest	lme4	nlme	MuMIn	vegan	dplR
17	Tree genetics	lme4	nlme	edgeR	ggplot2	WGCNA
18	Tree growth	nlme	lme4	dplR	MuMIn	ggplot2
19	Urban green space & health	lme4	vegan	car	MuMIn	nlme
20	Wildfires	lme4	vegan	nlme	MuMIn	car

and health (FAO and UNEP, 2020), it was reassuring to note that the number of forest ecology/forestry publications among the 13 journals we examined increased from 820 per year in 2010 to 2,354 in 2022; however, this increase varied across the four subject fields (Forest growth, Forest management, Forest use, and Forest monitoring) (Figs. 2, 3; Table S4). This variability also applied across topics, despite an absolute increase in the overall number of published articles relating to each topic. Specifically, the proportional representation of conceptual topics such as Biodiversity and ecosystem function increased while more applied topics, such as Forest regeneration mechanisms, comprised a smaller proportion of annual articles toward the end of the data collection period. This is important because while all conservation action should be informed by sound theory and a broad conceptual understanding of implications, it is equally important that those practical procedures used to apply forestry practices are also well conceived and researched (Huffaker and Messenger, 2012; MacDicken et al., 2015). Macroscale topics also ascended in relative importance within the corpus (e.g., Biodiversity & Ecosystem Function, increasing from 64 articles in 2010 to 255 in 2021), as did studies on the topics of direct anthropogenic threats to forests (e.g., Land use change; (Chaudhary et al., 2016), which increased from 19 articles in 2010 to 49 in 2021, and indirect threats (e.g., Climate change modelling (Keenan, 2015)), increasing from 19 articles in 2010 to 129 in 2021.

This increasing annual trend in the numbers of papers published relating to forest ecology/forestry is consistent with similar trends in ecology generally (Saravanan and Dominic, 2014), and in zoology (Zhao et al. submitted). This likely relates to the liberation of journals from printed volumes, and to online 'open access', which has given rise to 'mega-journals' with broad disciplinary scope (Wakeling et al., 2016), while simultaneously supporting a steady stream of submissions to dedicated forest ecology journals (Bojovic et al., 2014); all sustained by greater analytical power and superior statistical tools (Nakagawa and Santos, 2012).

4.2. Geographic patterns in forest ecology topics

Internationally, the majority of these publications was published by first authors based in the USA, followed by China, Canada, Germany, and Australia. In contrast, many developing countries had little or no footprint in the literature (Zhao et al., submitted); for example, several West and Central African nations, Central American nations, and various small island nations (articles from these countries may have been published in local journals that were not part of our corpus which acts as a constraint on inclusivity in our approach). This is despite forests being critically important to the ecosystems and local livelihoods in many of these countries (Fig. S4). However, the notable absence of the UK, which has proportionately little forest cover, from the top countries publishing forest ecology research articles suggests that the extent of national forest cover is an important co-factor affecting how nations invest research funding. Capacity to conduct research is also a function of educational standards in host countries and/or their ability to draw in international scientists to their universities and research institutes (Bai et al., 2009), which may lag behind in developing countries (Sanyal and Varghese, 2006). In counterpoint, other nations, such as the Russian Federation and Brazil, which both have globally significant forest cover, were not among those countries where studies originated from in our corpus.

The discontinuity between where research is conducted and where it is most needed was further illustrated by our finding that hotspot study areas that featured in our corpus were located predominantly in Europe (34.3%), North and South America (37.8%), and Asia (21.7%) (MacDicken, 2015). This highlights the critical need for a more comprehensive global overview of forest ecology/forestry research (Libman and Obydenkovay, 2014; Rezende et al., 2018).

4.3. Application of statistical techniques to forest ecology/forestry

Forest ecology and forestry research are becoming ever more data-driven, with an increasing dependence on statistical software to conduct tabulation, visualization, and modelling (Zou et al., 2019). Over the 16,952 journal articles we analysed, we found that the popularity of the R language for statistical computing (R Coreteam, 2021) increased substantially from 2010 to 2022 (Figs. 5 and S5). This trend was evident among all 13 journals, although to a lesser extent among articles relating to applied forestry topics. R was the dominant choice for statistical analyses in 19 of the 20 topics, with the exception of Urban green space & health. Consistent with our findings, Atkins et al. (2022) also reported that forestry and forest ecology research lags behind related fields such as ecology, biodiversity, and conservation research in the employment of open-source software solutions, specifically the R programming language. They found that while R was utilised in over 30% of ecology literature, it was used in less than 10% of forestry literature. Nevertheless, the temporal trend in the use of R statistical software in our corpus was positive, increasing from 29.5% in 2010 to over 70% by 2022. This is indicative of the degree to which more sophisticated analyses have permeated forest ecology and forestry, requiring superior computational and statistical resources. For example, statistical modelling has become a vital tool for wildfire management (Zhang et al., 2019; Pham et al., 2020) and for predicting the effects of climate change of forest ecosystems (McDowell et al., 2020; Zhang et al., 2022). Furthermore, it is vital that forest ecology and forestry research adopts non-proprietary, open-source software (Dhir and Tsai, 2017) to facilitate reproducibility and foster collaboration (Buck, 2015; Atkins et al., 2022).

The R programming language was first introduced in 2000 (Peng, 2022). Since then, as an open-source application, the range of packages available within the R domain has developed considerably (Hornik and The R Coreteam, 2022), aided by user-created resources (Wickham et al., 2023), with over 19,000 packages currently available (Bakker, 2023). Benefiting from the exchange of knowledge within the R-user community (Lai et al., 2019), the versatility and flexibility of R software increases the range of analyses that can be conducted within this single interface environment (Hesselbarth et al., 2021).

The packages used most frequently were 'nlme', 'lme4', 'vegan' and 'MuMIn' which are used to fit and elevate complex models, and are widely used for multivariate analysis in community ecology (Lai et al., 2019). For example, the 'vegan' package includes many different functions to deal with forest community statistics (Oksanen et al., 2020). The further application of these packages to forestry-specific statistics is vital. Similarly, packages dealing with specific topics such as hyperspectral or LiDAR also have potential to advance analyses (Atkins et al., 2022).

5. Conclusion

While the increase in forest ecology/forestry research publications over the last 12 years is encouraging, it is vital that interdisciplinary forest management actually benefits from this enhanced knowledge base (Börner et al., 2020). To achieve Aichi Biodiversity Target 11 (IUCN, 2019b), forest conservation and the rate of reafforestation must ultimately come to exceed the concerning the rate of forest loss, and the quality of global forests must be maintained or enhanced to support vital ecosystem functions (Di Sacco et al., 2021), biodiversity (Storch et al., 2020) and livelihood security (Kaur and Mittal, 2020). The alternative, that this wealth of forest research should cause 'paralysis by analysis' (Litvaj et al., 2022) through over-thinking issues, causing uncertainty, contradiction, and a lack of coherent policy direction, must be rigorously avoided. To these ends we endorse the four key policy principles proposed by Polasky et al. (2020): (i) to follow the strongest and most direct path between policy decisions and outcomes; (ii) to focus on finding

sufficient evidence supporting policy purpose; (iii) to prioritise no-regrets policies by avoiding options with controversial, uncertain, or immeasurable benefits; and (iv) to aim at getting the big picture right rather than getting lost in fine details.

Funding

This work was financially supported by the National Natural Science Foundation of China (31971541).

Availability of data and materials

The data and code that support the findings of this study are publicly available via the GitHub repository at https://github.com/Suorgund/Bibliometric-analysis-in-forest-ecology.

CRediT authorship contribution statement

Jin Zhao: Writing – original draft, Conceptualization. **Liyu Li:** Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jian Liu:** Visualization, Formal analysis, Data curation. **Yimei Yan:** Visualization, Formal analysis, Data curation. **Qian Wang:** Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Chris Newman:** Writing – review & editing, Writing – original draft. **Youbing Zhou:** Writing – review & editing, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to thank Qian Qian for their assistance in data collection, checking and visualization.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fecs.2024.100233.

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